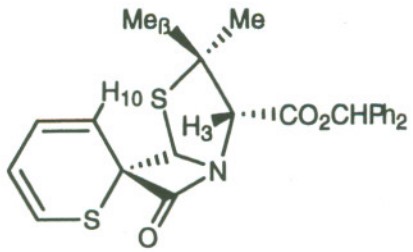
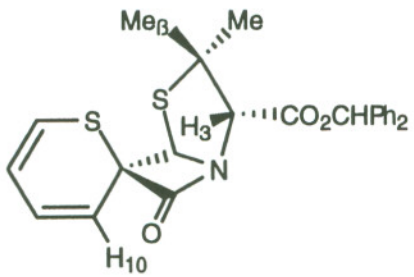
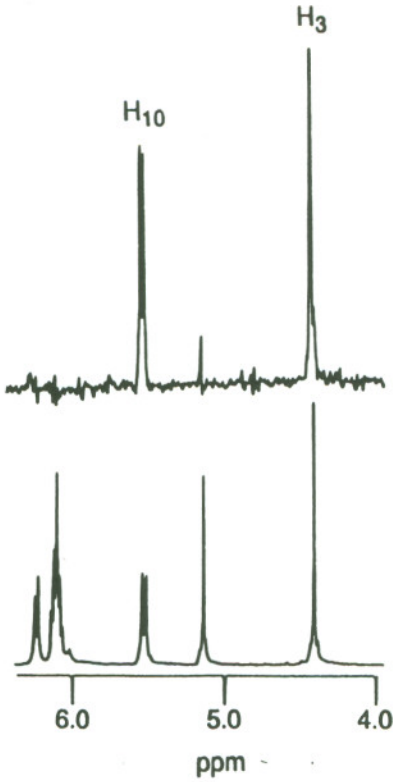
1) The following difference NOE spectra (top) resulted from irradiation of Me_β in one of the two penicillin derivatives shown, clearly identify which structure is correct



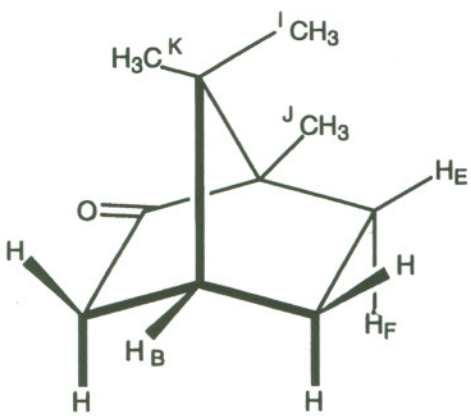
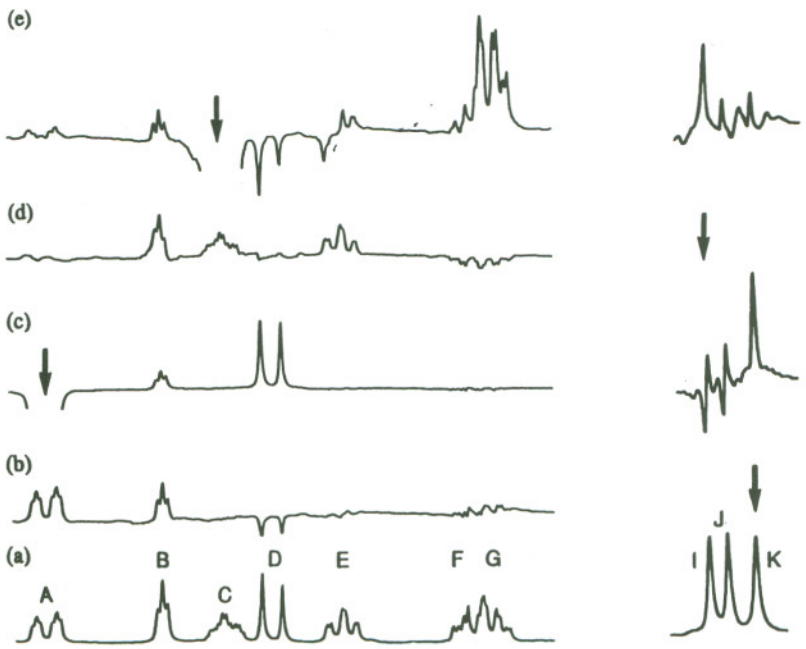
1



2



2) Given the following set of difference NOE spectra for **3**, identify the unlabeled protons in the structure as A, C, D, or G. The resonances irradiated (bold arrow) was as follows: (a) control spectra, (b) irradiation of K, (c) irradiation of A, (d) irradiation of I, (e) irradiation of C.



3